

SULAV GAIRE

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 [Sulav-Gaire](#) |  [SulavGaire](#)

Bhaktapur, Nepal

RESEARCH INTERESTS

My research interests focus on Electronics, Machine Learning, and IoT, with particular emphasis on mobile computing and cyber-physical systems. I am currently exploring Wi-Fi sensing in a personal project and aim to further develop my work in these fields.

EXPERIENCE

- **Nepal Academy of Science and Technology (NAST)**
Innovative Research Assistant Dec 2024 - Present
Khumaltar, Lalitpur, Nepal
 - Developing wireless sensor network for early warning system
 - Researching tomato picking robot using robotic arm and imitation learning.
 - Handling electronics aspects of projects like hydrogen fuel cell, GPS data logger.
- **ORION Space** 
Electronics Design Intern Feb 2022 - Aug 2022
Duwakot Mod, Bhaktapur
 - Researched LoRa-based sensor networks in UHF band
 - Engineered educational satellites (CanSat, CUBEK) used in global space training programs [20+ countries]
 - Developed web application for real-time LoRa data monitoring

EDUCATION

- **Institute of Engineering, Tribhuvan University** Nov 2019 – April 2024
Bachelor's Degree Pokhara, Nepal
 - Electronics, Communication and Information Engineering

PROJECTS

- **StaticSense: Static Object Detection using Wi-Fi Channel State Information** 2025
Tools: Resnet, Signal Processing, Esp32-WiFi-CSI 
 - Researched static object detection using Wi-Fi CSI, achieving 98.27 percent accuracy, addressing a challenge previously limited to dynamic activity sensing. Currently under review in International Conference on ICT and Photonics (ICTP), 2026
- **CUBEK & EDUCAN: Educational Satellites** 2022
Tools: Node.js, Arduino, PCB Design, LoRa 
 - Engineered 1U CubeSat kit with solar power systems and LoRa communication modules
 - Developed real-time dashboard and optimized PCB design, reducing production costs by 30%
- **Electronic Voting System: Biometric Authentication** 2024
Tools: Python, Flask, OpenCV, SHA256 Hashing 
 - Developed secure two-factor biometric voting system with fingerprint and facial verification
 - Achieved positive user ratings from 50 test participants with intuitive, interactive interface
- **Waste Classification Delta Arm** 2024
Tools: YOLOv8, OpenCV, Arduino 
 - Built AI-powered sorting robot achieving 94.1% precision, winning 1st place among 50 teams
 - Engineered end-to-end system for classifying waste into four categories with delta arm mechanics
- **High Altitude Weather Forecasting** 2023
Tools: LoRa, XGBoost, Python, Arduino 
 - Created cost-effective LoRa-based weather prediction system for remote mountainous regions
 - Implemented XGBoost algorithm to forecast patterns from sparse environmental sensor data

PUBLICATIONS

C=CONFERENCE, J=JOURNAL, P=PATENT, S=IN SUBMISSION, T=THESIS

- [C.1] Shiva Shrestha, Samman Shrestha, Prateek Paudel, Sudarshan Gurung, Sulav Gaire, Smita Adhikari (2024). **Deep Learning for Waste Management: Leveraging YOLO for Accurate Waste Classification**. In *Proceedings of the 15th IOE Graduate Conference*, Volume 15, May 2024. Institute of Engineering, Tribhuvan University. ISSN: 2350-8914 (Online), 2350-8906 (Print).

SKILLS

- **Programming Languages:** Python, MATLAB, JavaScript, Shell, C, C++
- **Web Technologies:** MERN Stack (MongoDB, Express.js, React, Node.js)
- **Database Systems:** MongoDB, MySQL
- **Data Science & Machine Learning:** Pandas, NumPy, Matplotlib, TensorFlow, PyTorch
- **Specialized Area:** IoT, Low-Power Wide-Area Networks (LPWAN), Embedded Systems
- **Other Tools & Technologies:** KiCad, LTspice, Figma, ROS2

HONORS AND AWARDS

- **Poster Presenter – WiFi-HAR: Activity Detection Using Channel State Information** 2025
International Conference on Science and Technology Information Management System: Practices and Experiences
◦ Organized by NAST and NAM S&T CENTRE)
- **National 2nd Runner-up – TECHENERGY Hackathon** 2024
Hitachi Energy 
◦ Recognized among top 3 out of 20+ national teams for software testing
- **1st Place – Project Expo (Delta Arm)** 2024
NebiFest 2024, IOE WRC
◦ Ranked 1st among 50+ projects for developing a low-cost robotic arm with real-time control
- **Regional 2nd Place – Robo-Soccer** 2021
CAN-INFOTECH Robotics Competition

LEADERSHIP EXPERIENCE

- **Senior Member – Robotics Club** 2020 – 2023
IOE, WRC
◦ Coordinated a 15-day robotics training for 100+ first-years freshers
◦ Led organizing team for in-house robotics competition; 20+ teams participated
◦ Mentored juniors in embedded systems and Arduino-based prototyping

ADDITIONAL INFORMATION

Languages: English (Proficient), Nepali (Native Speaker), Hindi (Intermediate)
Interests: Engineering, Art, Dance

:) *HAPPY ENGINEERING*